one sample

T

estimate

- population variance is unknown
- sample size less than 30
- population is normal

"S^2"

Z

- population variance is known
- sample size ≥ 30
  ≤ 30
- population is normal

"σ^2"

two sample

paired

dependendent

after - before

independent

* Variance unknown

unequal

equal

$$* \; S^2_{Pooled} = \frac{S_1^2[n_1-1] + S_2^2[n_2-1]}{n_1+n_2-2}$$

Hypo

left tail

- $H_0: \mu_1 = \mu_2 \rightarrow \mu_1 - \mu_2 = 0$
- $H_1: \mu_1 < \mu_2 \rightarrow \mu_1 - \mu_2 < 0$

right tail

- $H_0: \mu_1 = \mu_2 \Rightarrow \mu_1 - \mu_2 = \text{zero}$
- $H_1: \mu_1 > \mu_2 \Rightarrow \mu_1 - \mu_2 > \text{zero}$

two tail

- $H_0: \mu_1 = \mu_2 \Rightarrow \mu_1 - \mu_2 =$
- $H_1: \mu_1 \neq \mu_2 \Rightarrow \mu_1 - \mu_2 \neq$

baker two	baker one
$\sigma_2^2 = 25$	$\sigma_1^2 = 16$
$n_2 = 12$	$n_1 = 10$
mean = 75	mean = 80

point estimate

$$\bar{X}_1 - \bar{X}_2 = 80 - 75 = 5$$

$$\text{confidence interval} = [\bar{X}_1 - \bar{X}_2] \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$[80-75] \pm 1.96 \sqrt{\frac{16}{10} + \frac{25}{12}}$$

lower: 1.24 upper: 8.76

$$P[1.24 < \mu_1 - \mu_2 < 8.76] = 95\%$$

عشان ال zero مش من الرقم

reject H_0 ← • $H_0: \mu_1 - \mu_2 = \text{zero}$

accept ← • $H_1: \mu_1 - \mu_2 \neq \text{zero}$ → ادعاء المفتش

---

ex

male	femal
25	19
$\bar{X}_1 = 30$	$\bar{X}_2 = 25$
$S_1 = 5$	$S_2 = 4$

point estimate

$$\bar{X}_1 - \bar{X}_2 = 30 - 25 = 5$$

$$\text{confidence interval} = [\bar{X}_1 - \bar{X}_2] \pm T[\tfrac{\alpha}{2}, n_1 + n_2 - 2] * \sqrt{S_p^2 [\tfrac{1}{n_1} + \tfrac{1}{n_2}]}$$

$$[30-25] \pm 1.96 \sqrt{21.4 [\tfrac{1}{25} + \tfrac{1}{19}]}$$

lower: 2.15 upper: 7.85

$$P[2.15 < \mu_1 - \mu_2 < 7.85] = 95\%$$

← الستات اكثر طفاءه من الرجال • متوسط زمن خدمة الستات اقل من متوسط زمن الخدمة للرجال

متوسط زمن الرجال اكبر من متوسط زمن الستات

reject H_0 ← • $H_0: \mu_1 - \mu_2 = \text{zero}$ (ستات، رجال)

accept H_1 ← • $H_1: \mu_1 - \mu_2 > \text{zero}$ (رجال، ستات)